CODE2CONNECT HACKATHON | MODULE B

SentinelRisk

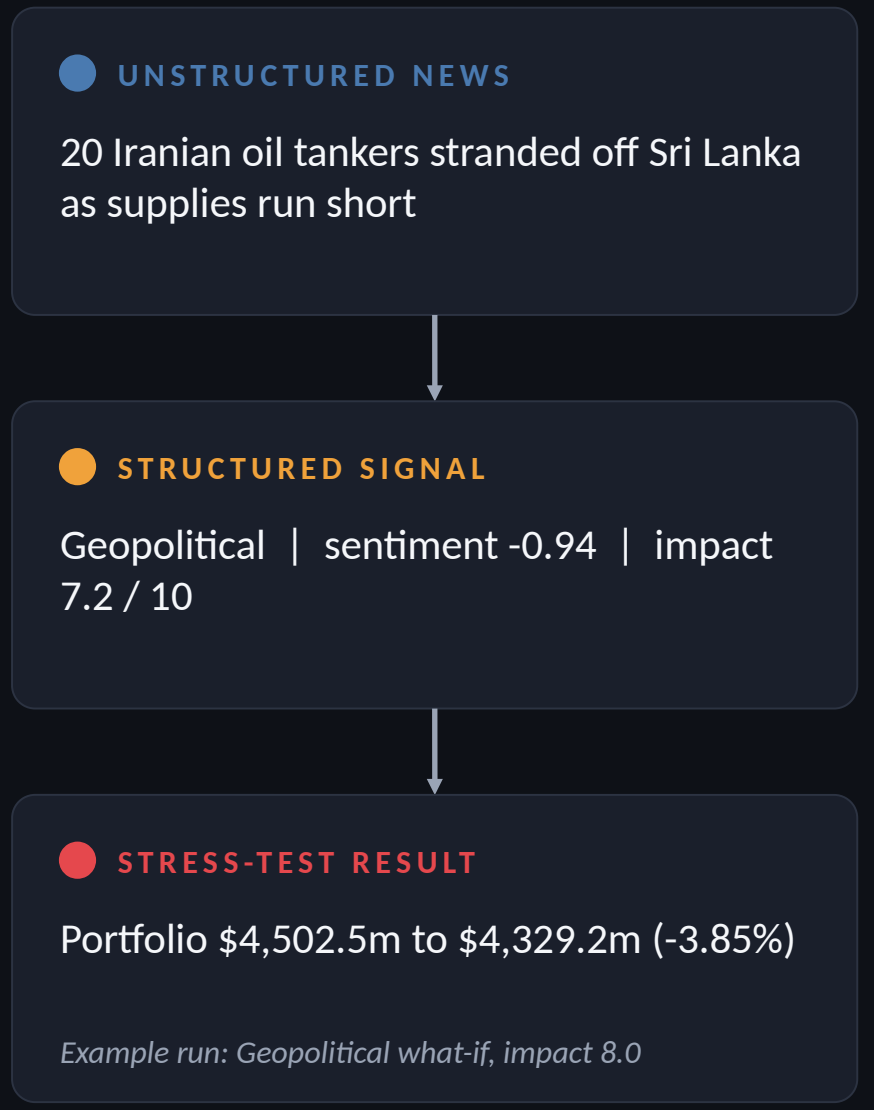
From Headlines to Stress Tests

An AI/NLP risk engine that turns unstructured financial news into structured signals, then automatically stress-tests a wholesale banking portfolio.

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GitHub: https://github.com/ayush-p-14k-18k/nitjalandhar_ayushparmar_code2connect



Problem and approach

The problem

Banks are flooded with headlines every day, but turning them into portfolio action is still manual. It is slow, inconsistent and hard to audit, so risk teams react after the market has moved.

5,026 headlines scored by the engine → **16** flagged high-impact (score above 7)

The signal is in the few events that matter

Our approach

1. Read the news

An NLP risk engine scores sentiment with FinBERT, classifies the event type and rates impact from 1 to 10.

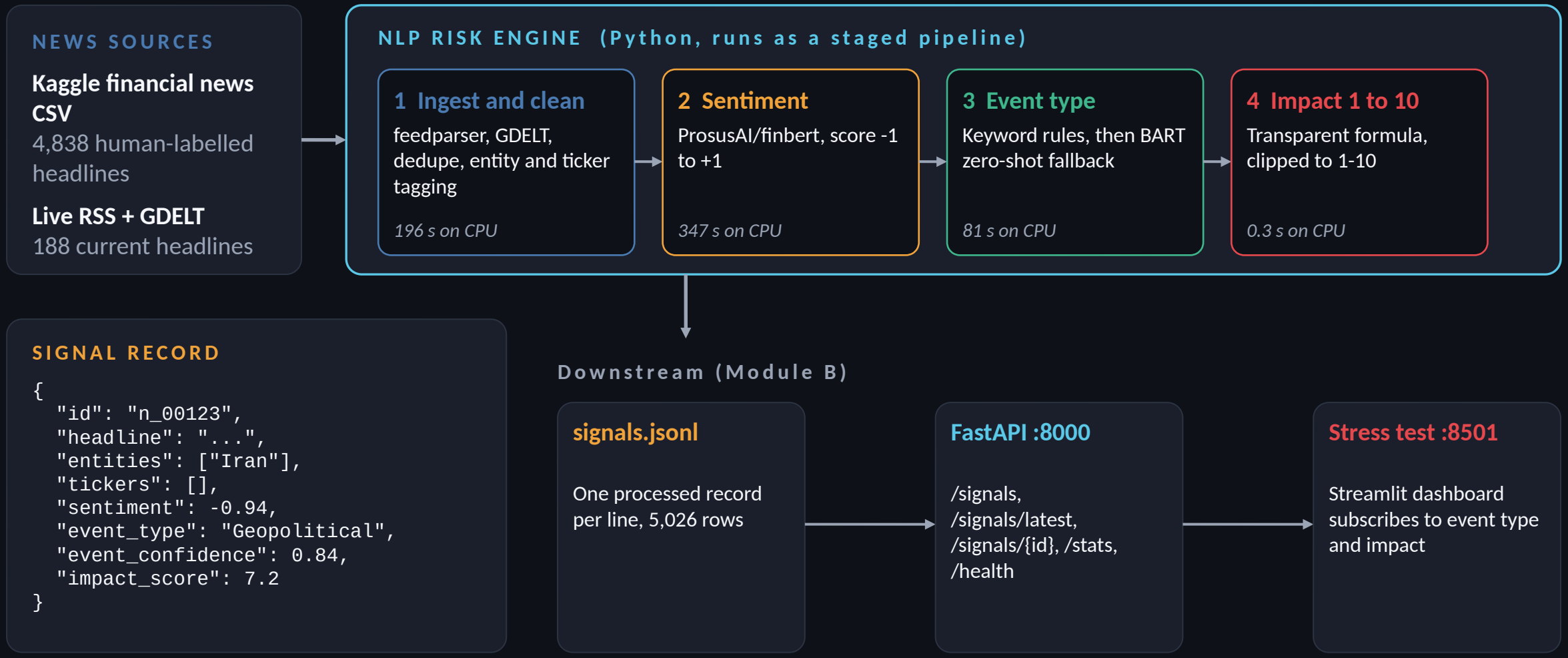
2. Publish structured signals

Every headline becomes a JSON record, served from a FastAPI service and a signals.jsonl file.

3. Stress test automatically

Module B listens for high-impact events and applies shocks to a synthetic portfolio, showing value before and after.

System design



Implementation highlights

Sentiment

P(positive) - P(negative)

- FinBERT (ProsusAI/finbert) gives one score from -1 to +1; label and confidence are kept
- Raw text goes in, with no stopword stripping, so negations survive
- Checked against 4,838 human labels: 88.9% accuracy, 0.97 recall on negative news

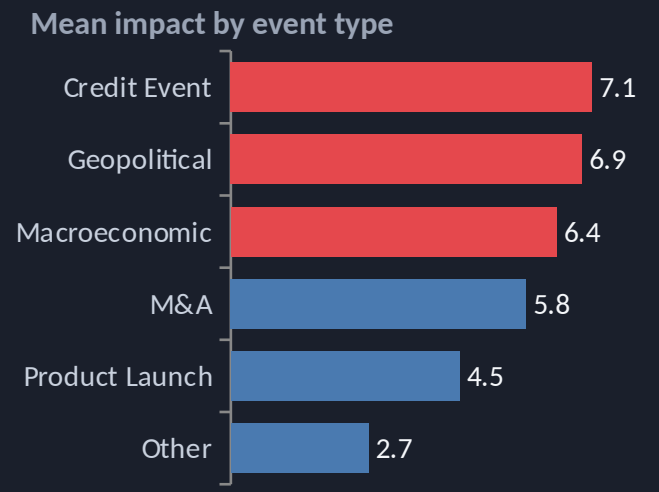
Event type

Rules first, model second

- Keyword rules are cheap and explainable: 469 headlines labelled
- BART zero-shot fallback labelled 132 more, 28% extra coverage over rules alone
- Five classes plus Other, each with a saved confidence

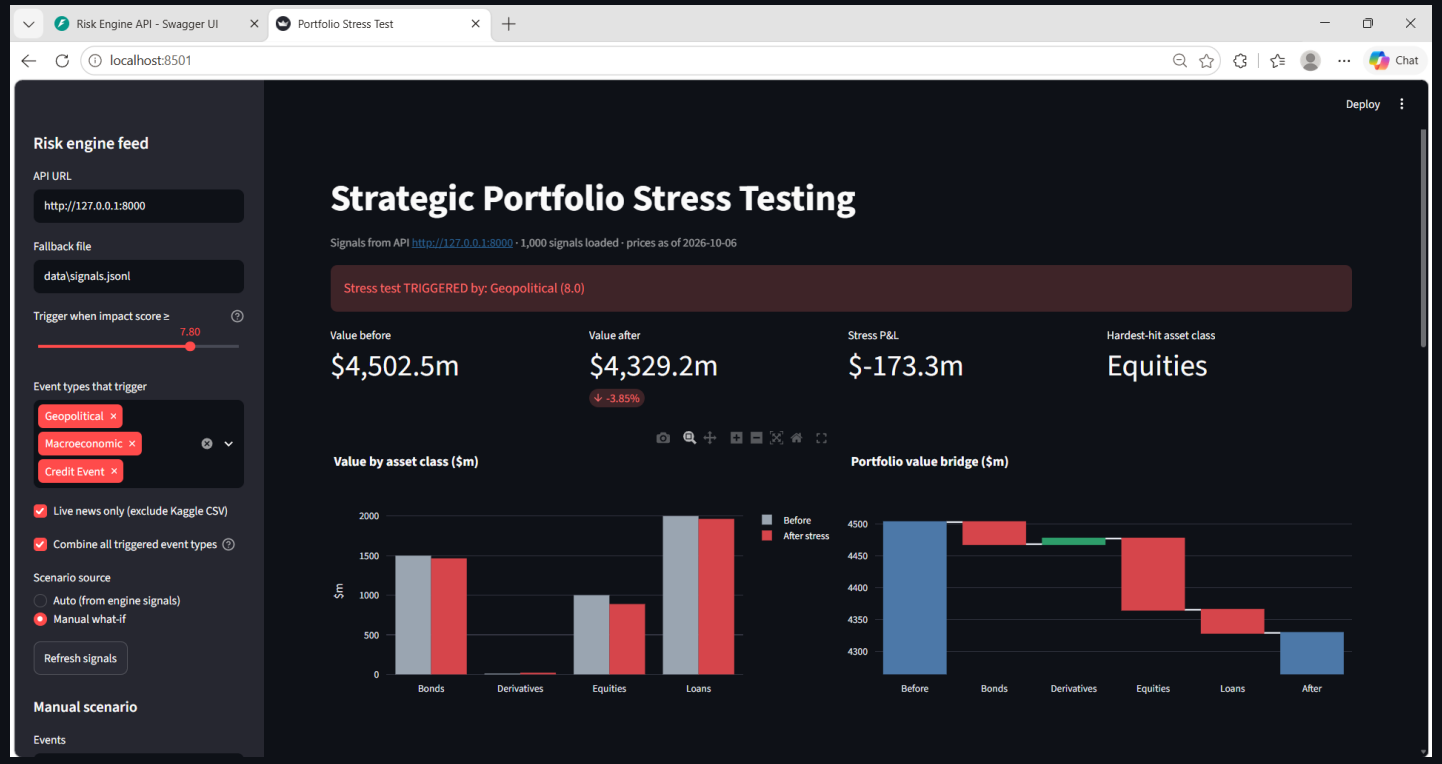
Impact score

```
base(event) + 2 x |sentiment|
+ intensity words + coverage
clipped to 1-10
```



Output: signals.jsonl + FastAPI endpoints /signals /signals/latest /signals/{id} /stats /health

Key results



Dashboard: a Geopolitical what-if (impact 8.0) triggers the stress test

- Equities take the biggest hit
- Derivatives gain, shown in green on the bridge
- Loans and bonds lose modestly

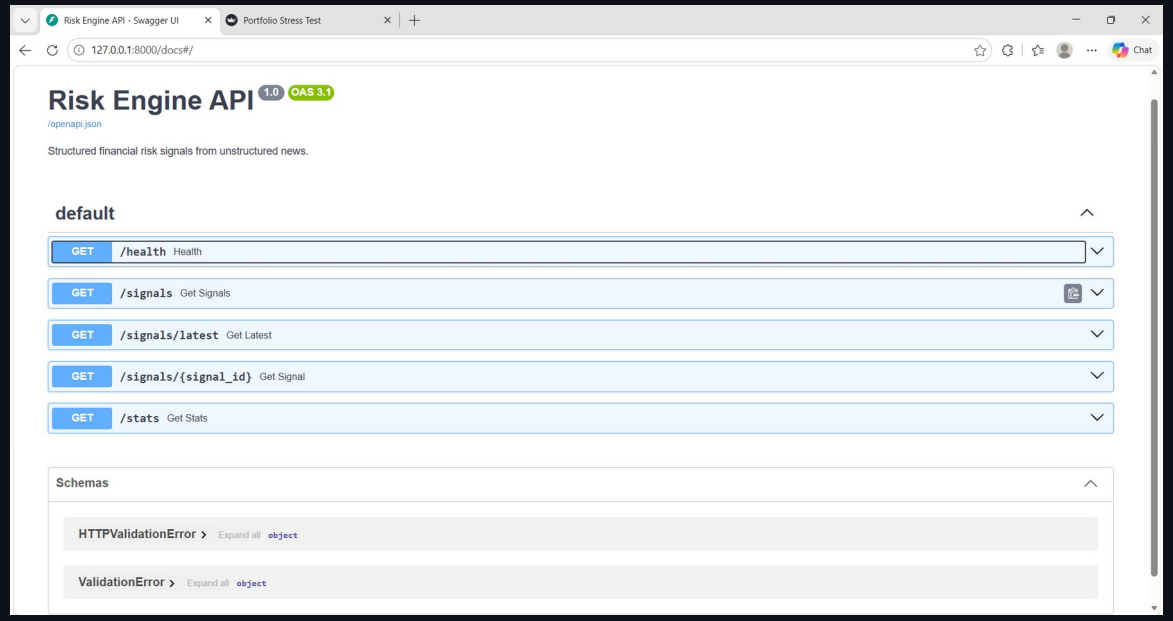
-3.85% — Portfolio value after stress: $4,502.5m to $4,329.2m (a $173.3m loss)

88.9% — FinBERT accuracy vs human labels on 4,838 headlines (macro-F1 0.882)

5,026 — Signals generated, 16 flagged high-impact (score above 7)

~10 min — Full pipeline on CPU; sentiment runs at about 14.5 headlines per second

Domain impact



Risk Engine API: any downstream system can subscribe

Headline in, scored signal out: 5,026 articles processed in about 10 minutes on a laptop CPU.

Early warning for risk teams

High-impact events surface automatically instead of waiting for a manual scan.

Explainable and auditable

Rules, a transparent impact formula and saved confidences make every score traceable.

Faster scenario analysis

Supports regulatory-style stress testing, with configurable thresholds and manual what-ifs.

Plug-and-play signals

A simple API means treasury, credit and trading tools can all consume the same feed.

Limitations and next steps

Limitations

- Impact score is a heuristic: no labelled market-severity data exists
- Kaggle headlines are company-level, so 94% fall under Other
- Keyword rules can misfire (a history headline tagged Geopolitical)
- Synthetic portfolio and fixed shocks; headline-level text only
- GDELT rate limits capped live volume at 188 headlines

Next steps

- Backtest impact scores against real market moves to calibrate weights
- Streaming ingestion for true real-time signals
- Map entities to actual exposures by sector, country and counterparty
- Learn event-specific shocks instead of fixed ones
- Multilingual news and full-article analysis

Try it

GitHub: https://github.com/ayush-p-14k-18k/nitjalandhar_ayushparmar_code2connect
Risk Engine API (local demo): http://127.0.0.1:8000/docs **Dashboard (local demo):** http://localhost:8501/